THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** an acid	Acid.
2	**(b)** 40 mL	Proportional.
3	**(a)** acidic	Acid rain.
4	**(d)** an acid	An acid neutralises the base.
5	**(b)** 30 mL	Doubling.
6	**(d)** it is a base used to relieve acidity	Carbonates react with acids.
7	**(a)** turns almost neutral	They neutralise.
8	**(b)** 500 kg	200 x 2.5.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Turmeric turns red (reddish brown) with bases, and soap solution is a base, so R explains A.
10	**(d)** A is false but R is true.	Acids turn blue litmus red and do not change red litmus; so A is false, but R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Lemon juice is acidic (1); It neutralises the base so turmeric turns back to yellow (1).	1+1
12	Use litmus or turmeric (1); Different colours appear (1).	1+1
13	Neither acid nor base (1); The effects of acid and base cancel out (1).	1+1
14	A mucus layer protects it (1); The acid is dilute (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ratio 10:20 (1); 60 is three times (1); 30 mL acid (1).	1+1+1
16	Some are more sensitive (1); Colour changes differ (1); Choose according to need (1).	1+1+1
17	$H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O$ (1); Acid and base (1); Salt and water (1).	1+1+1
	OR: Add slaked lime (1); Add wood ash (1); Add powdered limestone (1).	1+1+1
18	Neutralisation releases heat (1); It is exothermic (1); Salt and water form (1).	1+1+1
	OR: Add equal juice (1); Compare the red colour depth (1); More intense means more acidic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Hydrochloric acid (1). (ii) It is a base (an antacid) (1). (iii) Neutralisation (1). (iv) Lemon juice is acidic (it contains citric acid) (1), so it would add more acid to the stomach and make the burning worse instead of neutralising the extra acid (1).	1+1+1+2
20	Soil treatment (1); Antacids (1); Acid rain (1); Treating wastes (1); Principle: acid + base gives salt + water (1).	1+1+1+1+1
	OR: Natural indicators are safe (1); Cheap (1); Less precise (1); Lab indicators precise (1); Choice by need (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the acid was stronger or more concentrated	Excess acid.
2	**(b)** acid-carbonate reaction	CO2.
3	**(d)** $HCl + NaOH \longrightarrow NaCl + H_2O$	Neutralisation.
4	**(d)** acidic	Red is acid.
5	**(d)** vinegar	Acetic acid.
6	**(c)** aqueous $NaOH$	Sodium hydroxide.
7	**(c)** sugar solution	Neutral.
8	**(c)** a paste of baking soda and water	A base neutralises the acid.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the release of heat does not explain why salt and water form.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The stomach has hydrochloric acid; a base neutralises the extra acid, which explains the relief.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is basic (1); Bases feel soapy (1).	1+1
12	Sodium hydrogencarbonate (baking soda) (1); Mild base (1).	1+1
13	To avoid accidents and spills (1); So that nobody uses it by mistake (1).	1+1
14	It makes the water acidic (1); Fish and plants cannot survive in it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Baking soda is a base (1); Reacts with acid (1); Releases carbon dioxide (1).	1+1+1
16	Add lime or another base (1); Test until neutral (1); Then release (1).	1+1+1
17	Baking soda (1); It is a base (1); Others acidic (1).	1+1+1
	OR: Bacteria make acid from sugar (1); Acid dissolves enamel (1); Brush with basic paste (1).	1+1+1
18	Neutralises formic acid (1); Use litmus (1); No colour change (1).	1+1+1
	OR: Soap turns it red (1); Lemon turns it yellow again (1); Acid neutralises (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Dark pink (magenta) (1). (ii) Green (1). (iii) No colour change, because sugar solution is neutral (1). (iv) The liquid is a base, because only bases turn turmeric red (1); it will turn red litmus paper blue, as bases do (1).	1+1+1+2
20	Collect rain in clean glass (1); Test with litmus (1); Compare with distilled water (1); Repeat in several places (1); Conclude and report (1).	1+1+1+1+1
	OR: Use litmus (1); Compare with tap water (1); If acidic, inform adults (1); Do not drink (1); Treat water (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** evaporation	Salt remains.
2	**(d)** washing soda is basic	Base turns it green.
3	**(b)** making an acid stronger	Opposite use.
4	**(d)** dilute hydrochloric acid	Acid.
5	**(d)** acid and baking soda	Carbonates with acids.
6	**(c)** carbon dioxide	CO2.
7	**(b)** acid neutralises the base	Neutralisation.
8	**(a)** it is corrosive	Burns tissues.
9	**(c)** A is true but R is false.	The sting has formic acid and baking soda (a base) neutralises it; so A is true, R is false.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true properties, but the sour taste of acids does not explain the slippery feel of bases.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bases dissolve grease (1); Soapy (1).	1+1
12	$H_2SO_4 + Ca(OH)_2 \longrightarrow CaSO_4 + 2\,H_2O$ (1); Calcium sulphate and water (1).	1+1
13	Lime is basic (1); Reduces acidity (1).	1+1
14	Bacteria multiply faster (1); More lactic acid forms (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Deposit is carbonate (1); Acid reacts with it (1); Gas is released (1).	1+1+1
16	It does not change litmus (1); It is neither acid nor base (1); Pure water is neutral (1).	1+1+1
17	Acidic (1); Damage teeth (1); Add sugar (1).	1+1+1
	OR: Litmus unchanged (1); Turmeric unchanged (1); Compare with water (1).	1+1+1
18	Strong bases are corrosive (1); They can burn the skin (1); Gloves protect the hands (1).	1+1+1
	OR: Test with red cabbage (1); More green means more basic (1); Compare (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Acidic (1). (ii) Quicklime (a base) (1). (iii) Neutralisation (1). (iv) The factory should treat the acidic waste with a basic substance (1) so that the acid is neutralised and the water in the canal is not harmful to fish (1).	1+1+1+2
20	Chemical bonds change (1); Heat released (1); Controlled by slow addition (1); Strong acid and base must be mixed with care (1); Danger if strong acid and base mix (1).	1+1+1+1+1
	OR: Safe for skin (1); Effective cleaning (1); Too strong harms (1); Balanced formulas (1); Testing by manufacturers (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
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